

Last Time

- State Space
- Numerical Integration
- Energy
- Lyapunov stability
- Dynamics from Energy

Today

- Generalized Coordinates
- Work
- d'Alembert's Principle
- Euler-Lagrange Equation

Generalized Coordinates

- Anything that isn't Cartesian positions
- Example: polar coordinates
- Usually denoted by q_n
- If we have exactly n coordinates where $n = \# \text{ DOFs}$, call q_n "minimal coordinates"
- We usually choose q_n so that we can ignore constraints and only parameterize the DOFs

Work

* How does a force change the kinetic energy of a particle?

$$T = \frac{1}{2} m \vec{v} \cdot \vec{v} \Rightarrow \dot{T} = m \vec{v} \cdot \dot{\vec{v}}$$

- Plug in $\vec{F} = m\vec{\dot{v}} \Rightarrow \dot{v} = \frac{1}{m}\vec{F}$

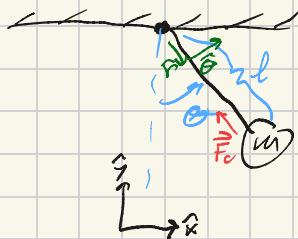
$$\Rightarrow \dot{T} = \vec{v} \cdot \vec{F}$$

$\Rightarrow$ Force only changes Kinetic energy when it is aligned with the velocity vector

- We'll capture this through the concept of "Work"

$$W = \Delta T = \int_{t_0}^{t_f} \dot{T} dt = \int_{t_0}^{t_f} \vec{F} \cdot \vec{v} dt = \int_{\vec{r}_0}^{\vec{r}_f} \vec{F} \cdot \underbrace{d\vec{r}}_{\text{infinitesimal displacement}}$$

- Let's think about the pendulum again:



$\downarrow \vec{g}$

$$\begin{aligned} x &= l \sin(\theta) \\ y &= l(1 - \cos(\theta)) \end{aligned}$$

$\downarrow$

$$\begin{aligned} \dot{x} &= l \cos(\theta) \dot{\theta} \\ \dot{y} &= l \sin(\theta) \dot{\theta} \end{aligned}$$

$$\vec{F}_c = \begin{bmatrix} -c \\ 0 \end{bmatrix}^T \begin{bmatrix} \hat{r} \\ \hat{\theta} \end{bmatrix} = \begin{bmatrix} -c \sin(\theta) \\ c \cos(\theta) \end{bmatrix}^T \begin{bmatrix} \hat{x} \\ \hat{y} \end{bmatrix}$$

$$\Rightarrow \vec{F}_c \cdot \vec{v} = \begin{bmatrix} -c \sin(\theta) \\ c \cos(\theta) \end{bmatrix}^T \begin{bmatrix} l \cos(\theta) \\ l \sin(\theta) \end{bmatrix} \dot{\theta} = cl \sin(\theta) \cos(\theta) - cl \sin(\theta) \cos(\theta) = 0$$

$\Rightarrow$ Constraint forces do no work!

(allowable $\vec{v}$ vectors must be perpendicular to constraints)

d'Alembert's Principle

- For a system of p particles:

$$\sum_{i=1}^p (\underbrace{\vec{F}_i - m_i \ddot{\vec{r}}_i}_{\text{"dynamics residual"}}) \cdot \underbrace{\delta \vec{r}_i}_{\text{"virtual displacement" satisfying constraints}} = 0$$

applied force (doesn't include constraints)

* Interpretation: 2 scenarios

1) Constraints satisfied $\Rightarrow \delta \vec{r} \neq 0 \Rightarrow \vec{F} - m\ddot{\vec{r}} = 0$
 $\Rightarrow$ Newton's 2nd law holds in "free directions"

2) Constraints violated $\Rightarrow \delta \vec{r} = 0 \Rightarrow \vec{F} - m\ddot{\vec{r}} \neq 0$
 $\Rightarrow$ Constraint force is active.

* Avoids explicitly computing constraint force $\vec{F}_C$

- To make sure $\delta \vec{r}$ satisfies constraints, we can write it in terms of minimal coordinates q :

$$\sum_{i=1}^n (\vec{F}_i - m_i \ddot{\vec{r}}_i) \cdot \frac{\partial \vec{r}}{\partial q} \delta q$$

- Applied to Pendulum:

$$\vec{F} = \begin{bmatrix} 0 \\ -mg \end{bmatrix}^T \begin{bmatrix} \hat{r} \\ \hat{\theta} \end{bmatrix} = \begin{bmatrix} mg \cos(\theta) \\ -mg \sin(\theta) \end{bmatrix}^T \begin{bmatrix} \hat{r} \\ \hat{\theta} \end{bmatrix}$$

$$\ddot{\vec{r}} = \begin{bmatrix} \ddot{r} - r\dot{\theta}^2 \\ r\ddot{\theta} + 2\dot{r}\dot{\theta} \end{bmatrix}^T \begin{bmatrix} \hat{r} \\ \hat{\theta} \end{bmatrix}$$

$$\delta \vec{r} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}^T \begin{bmatrix} \hat{r} \\ \hat{\theta} \end{bmatrix} \quad (\text{can only move in } \hat{\theta} \text{ direction})$$

$$\Rightarrow (\vec{F} - m\ddot{\vec{r}}) \cdot \delta \vec{r} = \left(\begin{bmatrix} mg \cos(\theta) \\ -mg \sin(\theta) \end{bmatrix} - m \begin{bmatrix} \ddot{r} - r\dot{\theta}^2 \\ r\ddot{\theta} + 2\dot{r}\dot{\theta} \end{bmatrix} \right)^T \begin{bmatrix} 0 \\ 1 \end{bmatrix} = 0$$

- 1st line is always zero $\Rightarrow$ ignore

$$\Rightarrow m l \ddot{\theta} + mg \sin(\theta) = 0 \Rightarrow \ddot{\theta} = -\frac{g}{l} \sin(\theta)$$

* Can ignore $\hat{r}$ direction, but still have to calculate $\ddot{\vec{r}}$ explicitly with change of coordinates.

Euler-Lagrange Equation

* We want to re-write d'Alembert's principle in terms of q and eliminate $\ddot{\vec{r}}$

- To keep notation light we'll focus on a single particle and express $\vec{F}$ and $\ddot{\vec{r}}$ in Cartesian components

$$(\vec{F} - m\ddot{\vec{r}}) \frac{\partial \vec{r}}{\partial q} \delta q = 0$$

$$\Rightarrow \left(\underbrace{\vec{F} \frac{\partial \vec{r}}{\partial q}}_{\text{"Generalized Force"}} - m\ddot{\vec{r}} \frac{\partial \vec{r}}{\partial q} \right) \delta q = 0$$

"Generalized Force"
Q

Two Identities:

$$1) \quad \dot{\vec{r}} = \frac{\partial \vec{r}}{\partial q} \dot{q} \Rightarrow \frac{\partial \dot{\vec{r}}}{\partial \dot{q}} = \frac{\partial \vec{r}}{\partial q}$$

$$2) \quad \frac{d}{dt} \left(\frac{\partial \vec{r}}{\partial q} \right) = \frac{d}{dt} \left(\frac{\partial}{\partial q} (r(t, q)) \right) = \frac{\partial}{\partial q} \left(\frac{d}{dt} (r(t, q)) \right) = \frac{\partial \dot{\vec{r}}}{\partial q}$$

Equality of mixed partials

$$\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial x}$$

- Note that:

$$\frac{d}{dt} \left(\dot{r} \frac{\partial r}{\partial \dot{q}} \right) = \ddot{r} \frac{\partial r}{\partial \dot{q}} + \dot{r} \frac{d}{dt} \left(\frac{\partial r}{\partial \dot{q}} \right)$$

$$\Rightarrow m \ddot{r} \frac{\partial r}{\partial \dot{q}} = \frac{d}{dt} \left(m \dot{r} \frac{\partial r}{\partial \dot{q}} \right) - m \dot{r} \frac{d}{dt} \left(\frac{\partial r}{\partial \dot{q}} \right)$$

- Using our 2 identities:

$$\begin{aligned} m \ddot{r} \frac{\partial r}{\partial \dot{q}} &= \frac{d}{dt} \left(m \dot{r} \frac{\partial r}{\partial \dot{q}} \right) - m \dot{r} \frac{\partial \dot{r}}{\partial \dot{q}} \\ &= \frac{d}{dt} \frac{\partial}{\partial \dot{q}} \left(\underbrace{\frac{1}{2} m \dot{r}^T \dot{r}}_{T \text{ (kinetic energy)}} \right) - \frac{\partial}{\partial \dot{q}} \left(\frac{1}{2} m \dot{r}^T \dot{r} \right) \end{aligned}$$

- Since T is a scalar, I can write it in any coordinates I want (i.e. q instead of r)

$$\Rightarrow \frac{d}{dt} \left(\frac{\partial T}{\partial \dot{q}} \right) - \frac{\partial T}{\partial q} = Q \leftarrow \text{generalized force}$$

- For conservative forces $Q_c = -\frac{\partial U}{\partial q}$ we can fold this in directly:

$$\frac{d}{dt} \left(\frac{\partial T}{\partial \dot{q}} \right) - \frac{\partial T}{\partial q} = \underbrace{-\frac{\partial U}{\partial q}}_{\text{conservative forces}} + Q \leftarrow \text{non-conservative forces}$$

- Define Lagrangian:

$$L = T - U \Rightarrow \boxed{\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}} \right) - \frac{\partial L}{\partial q} = Q}$$

* Notes :

- 1) This expression is coordinate invariant (q can be anything)
- 2) We don't have to compute $\ddot{\mathbf{r}}$ via coordinate transformations anymore
- 3) We just have to compute energy and generalized force in chosen coordinates.

- Pendulum one more time :

$$q = \theta, \quad T = \frac{1}{2} m l^2 \dot{\theta}^2, \quad U = m g l (1 - \cos(\theta))$$

$$\Rightarrow L = T - U = \frac{1}{2} m l^2 \dot{\theta}^2 - m g l (1 - \cos(\theta))$$

$$\frac{\partial L}{\partial \dot{\theta}} = m l^2 \dot{\theta}, \quad \frac{\partial L}{\partial \theta} = m g l \sin(\theta)$$

$$\Rightarrow \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\theta}} \right) - \frac{\partial L}{\partial \theta} = m l^2 \ddot{\theta} - m g l \sin(\theta) = 0$$

$$\Rightarrow \ddot{\theta} = \frac{g}{l} \sin(\theta)$$